Peer-Review Report

Manuscript: Chapter 19 — Missing Data in Mass Spectrometry: Mechanisms, Imputation Theory, and Sensitivity Auditing

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Reviewer Stance: Springer / Nature-style peer reviewer

Review Scope: Mathematical correctness of all 21 numbered equations; internal numerical consistency between narrative, R code, and exercise solutions; accuracy of attributed citations.

Severity Legend

CRITICAL	Blocks acceptance — must be fixed before publication.
MAJOR	Materially weakens the chapter; revision required.
MINOR	Editorial / clarification-level; non-blocking.

1. Executive Summary & Verdict

The chapter delivers a mathematically rigorous and pedagogically ambitious treatment of missing-data mechanisms in mass spectrometry, covering Rubin's MCAR / MAR / MNAR taxonomy, six imputation algorithms, Rubin's multiple-imputation pooling rules with the Barnard-Rubin (1999) small-sample degrees-of-freedom adjustment, Templ's Average Difference of Covariance Structure (ADCS) metric, and the imputation-free proDA likelihood of Ahlmann-Eltze and Huber (2020). The structure — roadmap, taxonomy, theory, hands-on R laboratory, Five Deadly Sins, exercises with worked solutions — is well-conceived, and the mathematical foundations are, on independent re-derivation, sound.

However, the chapter is recommended for Major Revision. Two Critical issues materially undermine the empirical demonstration and a narrative claim: (i) the Hybrid MAR/MNAR benchmark implementation degenerates to k-NN for every protein, producing a benchmark row byte-identical to the k-NN row; (ii) the chapter's universal claim that Little's MCAR test "almost universally rejects MCAR ($p < 10^{-4}$)" is contradicted by the actual R output in Listing 6, which reports $p = 0.184$. Five Major issues (R version mismatch, an unexplained 6× power drop for Rubin MI, an unjustified proDA flag, a typographical defect in Exercise 19.1, and an erroneous citation for Cox et al. 2014) require correction. Four Minor issues concern terminology, simulation transparency, and wording. All 21 numbered equations and all 22 numerical results in the four worked exercise solutions were independently re-derived and verified correct.

2. Strengths

The chapter's mathematical core is reliable. Rubin's three mechanism definitions (Eqs 19.2–19.4) are stated in their correct conditional-probability form, with the MAR conditional independence $P(M \mid Y_{obs}, Y_{mis}, \phi) = P(M \mid Y_{obs}, \phi)$ captured precisely. The four Rubin (1987) pooling formulas (Eqs 19.12–19.15) — pooled point estimate $\bar{Q}$, within-imputation variance $\bar{U}$, between-imputation variance B, and total pooled variance $T = \bar{U} + (1 + 1/M)\, B$ — are all correct, including the finite-imputation correction factor $(1 + 1/M)$.

The Barnard-Rubin (1999) small-sample degrees-of-freedom adjustment is also correctly derived. Eq 19.16 expresses $\nu_{obs} = [(\nu_{com}+1)/(\nu_{com}+3)] \times \nu_{com} \times (1 - r_m/(1+r_m))$, where $r_m = (1+1/M)\, B / \bar{U}$ is the relative variance increase due to non-response. The λ fraction of missing information appears here as $r_m/(1+r_m)$, which is algebraically equivalent to the standard $(1+1/M)\, B / T$ form. Eq 19.17 then combines $\nu_{old} = (M-1)(1 + r_m^{-1})^2$ with ν_{obs} via the harmonic-mean-like formula $\nu_{adj} = \nu_{old}\, \nu_{obs} / (\nu_{old} + \nu_{obs})$, matching the Barnard-Rubin (1999) paper. The proDA sigmoidal dropout (Eq 19.7) $P(M = 1 \mid y) = 1 / (1 + \exp((y - \rho)/\zeta))$ has the correct direction: high abundance → low missingness, low abundance → high missingness.

Templ's ADCS metric (Eq 19.21) is consistent: $(1/p) \times \| \Sigma_{imp} - \Sigma_{true} \|_F$ equals the square root of the mean squared per-element error, which is the RMS per-element covariance error (see Minor m4 for a wording nit). The pedagogical scaffolding — a four-checkpoint roadmap (Table 19.1), a mathematical classification of imputation methods (Table 19.2), the Five Deadly Sins, and a self-assessment diagnostic checklist (Table 19.3) — is genuinely useful for analytical chemists entering this literature.

3. Critical Issues

Two issues must be resolved before publication. Both are confined to the empirical demonstration and to a single narrative claim; neither affects the chapter's mathematical core.

CRITICAL **C1. The Hybrid MAR/MNAR benchmark row is byte-identical to the k-NN row — the segmentation never triggers.**

Location. Listing 2 (Hybrid segmented imputation), Table 19.2 row 5, Table 19.4 (Templ audit), and § 19.3.4 narrative.

Problem. Listing 2 sets the MNAR / MAR routing threshold as `is_mnar_prot <- mean_obs < 20.8`. The simulation generates per-cell intensities from $N(22, 1.2^2)$ (Listing 1, line `truth <- matrix(rnorm(p*n, mean=22, sd=1.2), nrow=p, ncol=n)`), so each protein's row mean across 20 samples is approximately $N(22, 1.2^2/20) = N(22, 0.27^2)$. The minimum observed row mean across 386 filtered proteins is therefore around $22 - 3.5 \times 0.27 \approx 21.1$ — already above 20.8.

Worse, MNAR censoring below LOD = 20.2 systematically removes low values, biasing observed row means upward. For the downregulated proteins (PROT_31..60, true Disease mean = 19.5), the Disease-condition observed values are those above 20.2; their conditional mean is approximately 21, raising the combined observed row mean to ≈ 22. Hence `is_mnar_prot[i]` evaluates to FALSE for every one of the 386 filtered proteins, and the Hybrid method uses k-NN for every protein — including the genuinely MNAR downregulated proteins.

Evidence. The differential-abundance benchmark (Listing 5) prints identical rows for k-NN and Hybrid:

Method	Sig	TP	FP	FDR%	Bias_Up	Bias_Down
k-NN (Donor Matching)	26	26	0	0.0	-0.244	1.700
Hybrid MAR/MNAR (Palstrøm)	26	26	0	0.0	-0.244	1.700

The Templ audit (Listing 6) likewise reports identical NRMSE = 1.38 and ADCS = 0.323 for both methods.

Impact. The chapter's central pedagogical claim — that hybrid segmented imputation "prevents the overestimation of absent features while preserving correlation structures among observed features" (§ 19.3.4) — is unsupported by the chapter's own simulation. Worse, Deadly Sin 3 (§ 19.5) prescribes hybrid MAR/MNAR as a remedy, but the very implementation demonstrated fails to exercise the MNAR branch.

Recommended fix. Either (a) raise the threshold to ≈ 22.0 — or, more principled, switch to a missingness-rate criterion (e.g. `is_mnar_prot <- rowMeans(is.na(obs_filtered)) > 0.30`) so that proteins with high dropout are routed to QRILC; or (b) acknowledge that the simulation parameters do not exercise the Hybrid pathway and remove the Hybrid row from the benchmark, while qualifying the prescription in Deadly Sin 3.

CRITICAL C2. Little's MCAR test result in Listing 6 (p = 0.184) contradicts the universal claim in §19.2.5.

Location. § 19.2.5 narrative; Listing 6 R output.

Problem. § 19.2.5 states, in the closing paragraph:

> "In high-throughput mass spectrometry, Little's test almost universally rejects MCAR ($p < 10^{-4}$) because instrument duty cycles and detection thresholds systematically induce abundance-dependent missingness."

Listing 6's actual R output, however, prints:

```
statistic = 33.4, df = 27, p.value = 0.184, missing.patterns = 7
```

The test fails to reject MCAR ($p = 0.184 \gg 0.05$). The contradiction arises because the 6-protein subset chosen (`obs_filtered[1:6, ]` — PROT_0001 to PROT_0006) consists of upregulated proteins with high abundance (true mean ≈ 22 in Control, 24 in Disease), for which MNAR dropout is rare and MCAR plausibly holds. The narrative's "almost universally rejects" claim may well be true on a representative low-abundance subset, but the chapter does not run the test on such a subset.

Recommended fix. Either (a) re-run Little's test on a stratified subset that includes low-abundance MNAR features (e.g. select 6 proteins with the highest missingness rates); or (b) qualify the universal claim: "Little's test rejects MCAR only when the analyzed block includes low-abundance MNAR features; on high-abundance subsets it may fail to reject MCAR even when MNAR dominates the full matrix."

4. Major Issues

MAJOR

M1. R version mismatch — listings say "[R 4.6 / Bioconductor]" but the reproducibility spec says "R 4.4.1".

Location. R Lab Listings 1–6 headers; § 19.4 "Computational Reproducibility Specification".

Problem. All six R Lab Listings are headed "[R 4.6 / Bioconductor]", while the chapter's final reproducibility paragraph states: "All analyses were executed under R version 4.4.1 (2024-06-14 ucrt) and Bioconductor 3.19." R 4.6 does not exist as of late 2025 (current R release is the 4.5.x series), and R 4.4.1 / Bioconductor 3.19 does exist. The two statements cannot both be true, and at least one is fictional. This is a reproducibility blocker: a reader trying to follow the labs cannot tell which environment to install.

Recommended fix. Standardise on a single, real release. Recommend "R 4.4.1 / Bioconductor 3.19" everywhere, including the listing headers, or update both to a current 4.5.x release after re-running the labs.

MAJOR

M2. Rubin MI finds only 5/60 true signals — a 6× power loss vs QRILC alone — without adequate explanation.

Location. Listing 5 benchmark table; Listing 3 top-5 Rubin summary; Figure 19.4E.

Problem. The benchmark shows:

Method	Sig	TP	FP	FDR%
QRILC alone (single imputation)	32	29	3	9.4
Rubin MI (M = 20, QRILC-based)	5	5	0	0.0

This 6× reduction is the chapter's central methodological point — single imputation is anti-conservative; proper variance accounting via Rubin's rules is more honest. But the explanation given is incomplete. The top-5 Rubin summary (Listing 3) reports Variance_Inflation of only 2–7%. With $(1 + 1/M) = 1.05$ and $B/\bar{U}$ small, the t-statistics shrink by a factor of $\sqrt{(1.05)} \approx 1.025$, which is far too small to explain a 6× drop in significance count.

The real driver is the Barnard-Rubin adjusted degrees of freedom: $\nu_{adj} \approx 15$ (Listing 3 top-5) versus limma's eBayes-moderated residual df (effectively very large). At $\alpha = 0.05$ two-sided, the critical t-value rises from 1.96 (df → ∞) to 2.13 (df = 15), and at typical $t \approx 3$ the two-sided p-value roughly doubles (from 0.003 to 0.009). After Benjamini-Hochberg adjustment across 386 tests, this doubling pushes borderline proteins out of significance. The chapter should make this mechanism explicit, otherwise a reader could plausibly suspect a bug in the Rubin pooling implementation.

Recommended fix. Add a paragraph after the benchmark table explaining that the 6× power loss is driven primarily by the Barnard-Rubin df adjustment ($\nu_{adj} \approx 15$ in this n = 20 cohort), not by the (1 + 1/M) variance inflation factor; and noting that QRILC-alone's 32 "discoveries" are almost certainly inflated by limma's eBayes df moderation, which has no analog for single-imputed data.

MAJOR M3. proDA called with moderate_location = FALSE without justification; biases the comparison.

Location. Listing 4, first line: `fit_proda <- proDA(obs_filtered, design = design, moderate_location = FALSE)`.

Problem. The proDA default moderates both location and variance (analogous to limma's eBayes). Disabling location moderation makes proDA's logFC estimates noisier and reduces its statistical power, biasing the comparison in favour of the imputation methods. The chapter does not justify this choice.

Recommended fix. If the rationale is to make proDA comparable to limma + impute (which does not moderate location), say so explicitly in the Listing caption. Otherwise, use the proDA default (`moderate_location = TRUE`) and re-run the benchmark.

MAJOR M4. Exercise 19.1 problem statement has missing leading digits in the Q■ vector.

Location. § 19.7, Exercise 19.1 problem statement.

Problem. The problem statement prints:

```
Q^ = (1.80, .10,.95, .05,.90)ᵀ
```

The leading digits "2", "1", "2", "1" are missing from the 2nd through 5th entries — apparently a typographical artifact where the leading characters were dropped during typesetting. The worked solution uses the intended values (1.80, 2.10, 1.95, 2.05, 1.90), which the student cannot recover from the problem as printed.

Recommended fix. Re-typeset the Q⊠ vector with explicit leading digits: `(1.80, 2.10, 1.95, 2.05, 1.90)ᵀ`.

MAJOR M5. Citation [9] (Cox et al. 2014, MaxLFQ) has multiple author errors.

Location. Bibliography entry [9].

Problem. The chapter lists: "Cox, J., Hein, M. Y., Luber, C. A., Parida, I., Cox, J., & Mann, M. (2014)." Three errors are present:

1. "Parida, I." should be "Paron, I." — the third author's surname is Paron, not Parida.

2. "Cox, J." is listed as both first and fifth author. Jürgen Cox is the first author only; the fifth author should be Nagaraj, N.

3. The author "Nagaraj, N." is missing entirely.

Recommended fix. Replace with: "Cox, J., Hein, M. Y., Luber, C. A., Paron, I., Nagaraj, N., & Mann, M. (2014). Accurate proteome-wide label-free quantification by delayed normalization and maximal peptide ratio extraction, termed MaxLFQ. Molecular & Cellular Proteomics, 13(9), 2513–2526."

5. Minor Issues

MINOR

m1. "Variance_Inflation" column header is the Relative Variance Increase, not the Variance Inflation Factor.

Location. Listing 3, `Variance_Inflation = round((T_var - U_bar) / U_bar * 100, 1)`.

This quantity is the relative variance increase (RVI) due to non-response, i.e. the percentage by which Rubin's total variance T exceeds the naive within-imputation variance $\bar{U}$. It is unrelated to the Variance Inflation Factor (VIF) of multicollinearity diagnostics in regression. Readers familiar with VIFs may be misled.

Recommended fix. Rename to "RVI (%)" or "Relative Variance Increase".

MINOR

m2. Figure 19.1A's LOD = 20.2 is the simulation's true parameter, not an estimated diagnostic.

Location. Figure 19.1A caption.

The caption reads: "The negative slope (γ□□ = −0.358, p = 1.78 × 10□□) establishes abundance-dependent MNAR left-censoring below the limit of detection (LOD = 20.2)." In a real mass spectrometry dataset, the LOD is unknown and must be estimated from blank / sample transitions; the LOD = 20.2 here is the simulation's dropout inflection point (ρ in Eq 19.7), not a value recovered from the data. The caption should clarify that this is a synthetic-data artifact, so readers do not infer that LOD can be read off the logistic curve in practice.

MINOR

m3. The simulation uses a single ρ for all samples; Eq 19.7 uses per-sample ρ■.

Location. Eq 19.7 vs Listing 1 line `p_mnar <- plogis(-(truth - 20.2) / 0.8)`.

Eq 19.7 specifies the proDA dropout probability with sample-specific parameters ρ_j and ζ_j. The simulation uses a single pair (ρ = 20.2, ζ = 0.8) for all samples, which is a reasonable simplification for demonstration but should be noted — in real DDA cohorts the LOD varies with instrument sensitivity per run, and proDA's core strength is its ability to estimate per-sample ρ_j and ζ_j.

MINOR

m4. ADCS expanded form should clarify it is the RMS per-element error, not the arithmetic mean.

Location. Eq 19.21 and § 19.3.6 narrative.

The chapter states: "The ADCS metric measures the average Euclidean error per element of the p × p covariance matrix." However, the expanded form $\sqrt{((1/p^2)\ \Sigma_j\ \Sigma_k\ (\Delta_{jk})^2)}$ is the root-mean-square per-element error (each Δ_{jk} enters squared, then square-rooted), not the arithmetic mean per-element error $(1/p^2)\ \Sigma\ |\Delta_{jk}|$. The two diverge substantially when Δ_{jk} has non-uniform magnitudes. Recommended fix: reword as "the RMS per-element error of the p × p covariance matrix."

6. Equation-by-Equation Verification

All 21 numbered equations in the chapter were independently re-derived against their cited primary sources. The table below summarises the audit.

Eq.	Description	Verdict	Primary Source
19.1	Missingness indicator $M_{ij} \in \{0,1\}$	Correct	Trivial
19.2–19.4	Rubin MCAR / MAR / MNAR conditional-probability definitions	Correct	Rubin (1976)
19.5	Logistic missingness diagnosis logit P(M=1 | ☒$_{obs}$)	Correct	Standard logistic regression
19.6	Little's MCAR multivariate omnibus statistic d^2_{MCAR}	Correct	Little (1988); Templ (2023)
19.7	proDA sigmoidal dropout P(M=1 | y)	Correct	Ahlmann-Eltze & Huber (2020)
19.8	MinDet deterministic substitution $y^*_{ij} = \min_k(y_{kj})$	Correct	Standard (Lazar et al. 2016)
19.9	Perseus left-shifted Gaussian $y^* \sim N(\mu - \delta\sigma, (\omega\sigma)^2)$	Correct	Tyanova et al. (2016)
19.10	k-NN distance-weighted donor matching	Correct	Hastie et al. (2001)
19.11	Hybrid segmented imputation (QRILC if MNAR, k-NN if MAR)	Formula correct; see Critical C1 for implementation bug	Palstrøm et al. (2020)
19.12	Rubin pooled point estimate $\bar{Q} = (1/M) \Sigma$ Q☒$_m$	Correct	Rubin (1987)
19.13	Within-imputation variance $\bar{U} = (1/M) \Sigma \hat{U}_m$	Correct	Rubin (1987)
19.14	Between-imputation variance $B = (1/(M-1)) \Sigma$ (Q☒$_m - \bar{Q})^2$	Correct	Rubin (1987)
19.15	Total pooled variance $T = \bar{U} + (1 + 1/M) B$	Correct	Rubin (1987)
19.16	Barnard-Rubin ν_{obs} small-sample df	Correct	Barnard & Rubin (1999)
19.17	Barnard-Rubin $\nu_{adj} = \nu_{old}\nu_{obs} / (\nu_{old} + \nu_{obs})$	Correct	Barnard & Rubin (1999)
19.18	RMSE on missing entries	Correct	Standard
19.19	NRMSE = RMSE / $\sigma(Y^{true}_{mis})$	Correct	Templ (2023)
19.20	MAE = $(1/\|M\|) \Sigma \|y^*_{ij} - y^{true}_{ij}\|$	Correct	Standard
19.21	Templ ADCS = $(1/p) \| \Sigma_{imp} - \Sigma_{true} \|_F$	Correct; see Minor m4 for wording	Templ (2023)

7. Exercise Solutions Audit

All four worked solutions (§ 19.9) were independently re-derived. Every numerical result — 22 in total across the four exercises — matches the chapter's reported values to the published precision.

Ex.	Sub-part	Reported	Re-derived	Match
19.1	$\bar{Q}$ (pooled point estimate)	1.960	1.960	✓

Ex.	Sub-part	Reported	Re-derived	Match
19.1	$\bar{U}$ (within-imputation variance)	0.0412	0.0412	✓
19.1	B (between-imputation variance)	0.01425	0.01425	✓
19.1	T (total pooled variance)	0.0583	0.0583	✓
19.1	$t = \bar{Q} / \sqrt{T}$	8.118	8.117	✓
19.1	r_m (relative variance increase)	0.4150	0.4150	✓
19.1	ν_{old} (Rubin asymptotic df)	46.50	46.50	✓
19.1	ν_{obs} (small-sample df)	11.51	11.51	✓
19.1	ν_{adj} (Barnard-Rubin df)	9.23	9.23	✓
19.2	$\mu_{imp} = \mu\boxtimes - \delta\sigma\boxtimes$	18.40	18.40	✓
19.2	$\sigma_{imp} = \omega\sigma\boxtimes$	0.60	0.60	✓
19.2	LOD (15th percentile)	19.927	19.927	✓
19.2	$P(Y_{imp} > \text{LOD})$	0.0055	0.0055	✓
19.3	$y^*_{Dis} = \boxtimes_{donors}$	21.5	21.5	✓
19.3	$\Delta_{true} = \mu_{Dis} - \mu_{Ctrl}$	−2.5	−2.5	✓
19.3	$\Delta_{knn} = y^*_{Dis} - \mu_{Ctrl}$	+1.5	+1.5	✓
19.3	$\text{Bias} = \Delta_{knn} - \Delta_{true}$	+4.0	+4.0	✓
19.4	$\| D_A \|_F$	1.000	1.000	✓
19.4	$ADCS_A = (1/3) \times 1.000$	0.333	0.333	✓
19.4	$\| D_B \|_F$	0.0755	0.0755	✓
19.4	$ADCS_B = (1/3) \times 0.0755$	0.0252	0.0252	✓
19.4	Ratio $ADCS_A / ADCS_B$	13.2×	13.21×	✓

Note on the +1.700 vs +4.0 attenuation discrepancy. § 19.3.3 narrative states "k-NN creates an attenuation bias of +1.700 log2 fold-change on downregulated features" (the empirical benchmark mean over 30 downregulated proteins), while Exercise 19.3's worked solution reports Bias = +4.0 log2 units (a hand-worked worst-case scenario with parameters μ_{Dis} = 17.5, μ_{Ctrl} = 20.0, $\boxtimes_{donors}$ = 21.5). These are not contradictory — one is the empirical mean across 30 proteins, the other is a single hand-worked example with extreme parameters. Adding a one-sentence clarification in § 19.3.3 to this effect would prevent reader confusion.

8. Recommendation

Verdict: Major Revision.

Required before acceptance:

C1. Fix the Hybrid threshold so the segmentation actually routes MNAR proteins to QRILC, OR remove the Hybrid row from the benchmark and qualify the prescription in Deadly Sin 3.

C2. Reconcile the Little's MCAR narrative (§ 19.2.5) with the actual Listing 6 output (p = 0.184). Either re-run Little's test on a stratified low-abundance subset, or qualify the "almost universally rejects" claim.

M1. Unify R version references to a single, real release (recommend "R 4.4.1 / Bioconductor 3.19" everywhere, or update both to a current 4.5.x release after re-running the labs).

M2. Add a paragraph explaining that the Rubin MI 6× power drop is driven primarily by the Barnard-Rubin $\nu_{adj} \approx 15$ (not by the (1 + 1/M) variance inflation factor), and that QRILC-alone's 32 "discoveries" are inflated by limma's eBayes df moderation.

M3. Justify or default the proDA `moderate_location` argument.

M4. Re-typeset Exercise 19.1's Q⊠ vector with explicit leading digits.

M5. Correct citation [9]: replace "Parida, I." with "Paron, I."; insert "Nagaraj, N."; remove the duplicate "Cox, J." entry.

Recommended (non-blocking):

m1. Rename the `Variance_Inflation` column to "RVI (%)" or "Relative Variance Increase".

m2. Clarify in the Figure 19.1A caption that LOD = 20.2 is the simulation's true parameter, not a recoverable diagnostic in real data.

m3. Note in Listing 1 that the simulation uses a single ρ for all samples (simplification of Eq 19.7's per-sample ρ_j).

m4. Reword the ADCS description in Eq 19.21 and § 19.3.6 as "RMS per-element error of the p × p covariance matrix" rather than "average Euclidean error per element".

The chapter's mathematical foundations are sound; all 21 numbered equations and all 22 exercise-solution numbers verify correct. The two Critical issues are confined to the empirical demonstration (the simulation's MNAR/MAR routing in Listing 2) and to a single narrative contradiction with the chapter's own R output (§ 19.2.5 vs Listing 6). Once these are addressed, the chapter will be a strong, mathematically rigorous, and pedagogically valuable contribution to the volume.